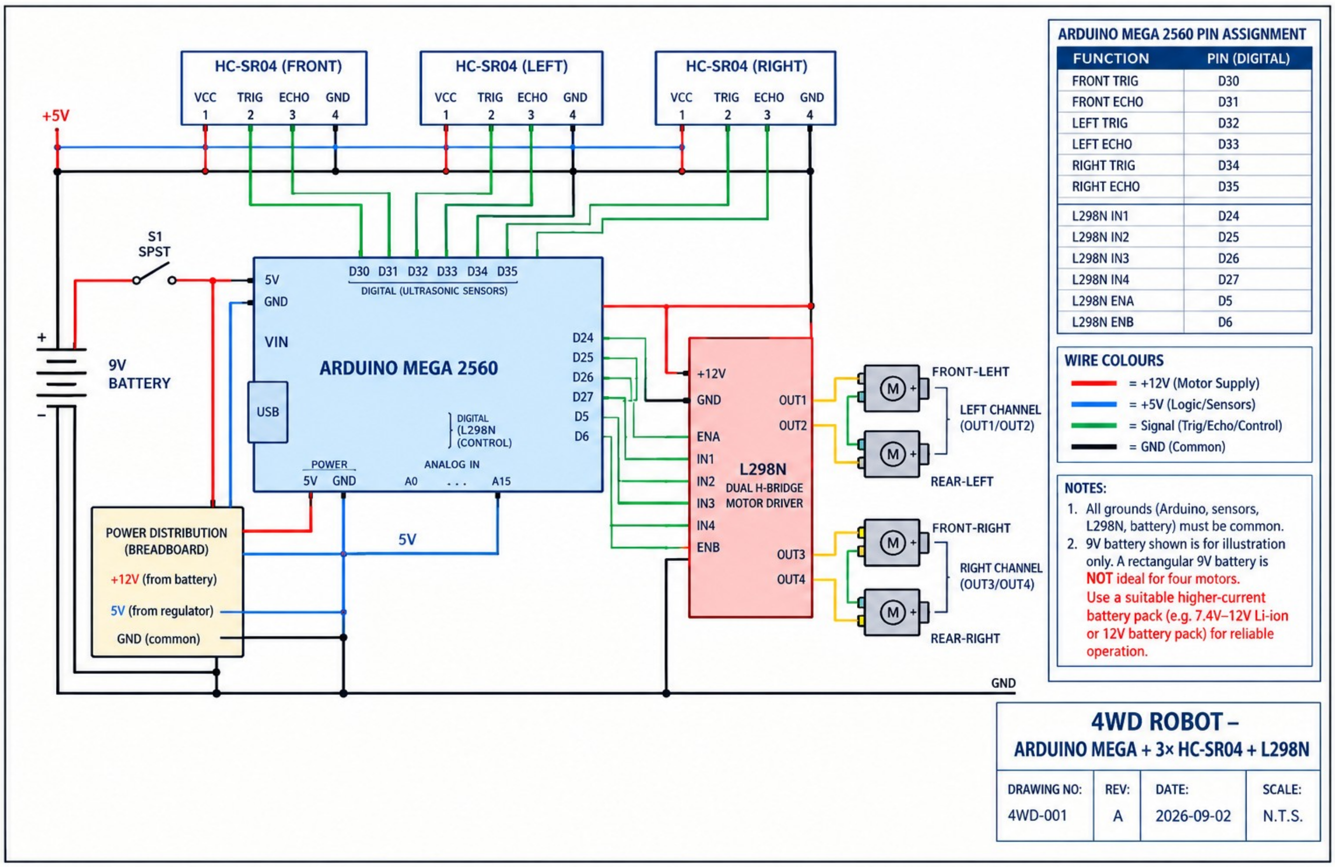
ROBO RUMBLE

AUTONOMOUS RACING

Holistic Build Document



PROJECT BASELINE

Autonomous 4WD racing platform using Arduino Mega 2560, three HC-SR04 ultrasonic sensors, an L298N dual H-bridge, and four geared DC motors. The supplied 9V PP3 battery is retained only as the reference-circuit item; the finished racer should use a higher-current rechargeable motor battery.

Prepared for Robo Rumble / Robo Grand Prix — Autonomous Racing

Competition date in project brief: 12 September 2026

1. Executive Summary

This build document defines the technical baseline for an autonomous racing robot intended for the Robo Rumble / Robo Grand Prix autonomous racing category. The objective is not simply to make a 4WD vehicle move, but to produce a compact, repeatable and serviceable platform that can sense its surroundings, make movement decisions without human steering, and convert those decisions into coordinated left/right motor commands.

The design uses an Arduino Mega 2560 as the central controller, three HC-SR04 ultrasonic sensors for front/left/right environment awareness, an L298N dual H-bridge for four-motor drive, and four geared DC motors arranged as two left motors and two right motors. The mechanical and electrical architecture is intentionally modular so that the control logic can be tuned without redesigning the full chassis.

- Primary race objective: maximize controllable speed while avoiding collisions and preserving a stable trajectory.
- Autonomy objective: all drive decisions are generated from on-board sensing and programmed logic; no steering input is required during the run.
- Engineering objective: remain inside the supplied category limits while keeping the center of mass low, wiring protected, and service access simple.
- Build objective: provide a clear verification trail from requirements -> wiring -> software -> testing -> competition readiness.

DESIGN STATUS

The document is written as Revision A using the circuit shown in the supplied reference image and the procurement BOM previously prepared. Physical prototype dimensions, exact chassis mass, motor current under load, and measured race speed must be verified on the assembled robot before a final competition release.

2. Background and Design Need

2.1 Problem Statement

In an autonomous race, the robot must react quickly enough to changes in path geometry or obstacles while maintaining traction and directional control. A manually driven 4WD platform can compensate for errors using a human operator, but an autonomous racer has to detect conditions, interpret them, and select a response inside a limited control cycle.

The technical challenge is therefore a closed-loop systems problem: sensing must be reliable, the controller must convert distance information into deterministic decisions, and the motor driver must apply those commands with enough electrical margin to avoid stalls, brownouts or uncontrolled resets.

2.2 Design Philosophy

- Keep the sensing geometry simple: one forward sensor establishes the primary collision-avoidance decision and two side sensors provide directional context.
- Treat left and right motor pairs as two drive channels. Differential speed between the channels provides steering without a conventional steering servo.
- Separate logic power from motor power through regulated supply and common ground architecture.
- Prefer deterministic state logic over unpredictable behavior. The robot should have explicit states such as READY, SEARCH, DRIVE, EVADE-LEFT, EVADE-RIGHT and SAFE-STOP.

- Design for debugging: named signals, documented pin assignments, accessible power switch and protected wiring.

3. Competition Category Constraints and Compliance

The following constraints are treated as the governing category requirements supplied for the Robo Rumble / Robo Grand Prix autonomous racing entry. The compliance status should be rechecked against the organizer's final technical inspection checklist immediately before scrutineering.

Constraint	Requirement	Design Response	Status	Verification
Footprint	Maximum 50 cm × 50 cm	Mechanical packaging must remain within the maximum footprint.	PASS BY DESIGN INTENT	Measure the complete outer envelope including sensors and wiring before inspection.
Mass	Maximum 5 kg	All chassis, electronics, battery and fasteners contribute to the competition mass.	TO VERIFY	Weigh the complete competition configuration.
Autonomy	Fully automatic	No manual steering/drive input during the race.	PASS BY ARCHITECTURE	Remove or disable any manual drive controls from the competition build.
Power	Battery only	Robot operates from onboard battery energy.	PASS BY ARCHITECTURE	Final battery must be onboard and isolated by the main switch.
Controller platform	Arduino / Raspberry Pi / STM accepted	The controller must belong to the accepted platform family.	PASS	Arduino Mega 2560 is used.
Category	Autonomous Racing / Grand Prix	Design emphasis is on autonomous navigation, speed, stability and repeatability.	PASS	Control strategy is closed-loop and sensor-driven.

INSPECTION GATE

A "PASS BY DESIGN INTENT" entry is not a measured certification. Final acceptance requires physical inspection of dimensions, mass, power system, wiring security and autonomous operation.

4. System Requirements

ID	Requirement	Meaning	Acceptance intent
FR-01	Sense environment	Measure front, left and right distances using HC-SR04 sensors.	Three sensor channels shall be sampled repeatedly during autonomous operation.
FR-02	Drive forward/reverse	Command coordinated motion of four motors through the L298N.	Left pair and right pair shall respond as two independently controlled drive channels.
FR-03	Steer autonomously	Change heading using differential left/right wheel speed.	Controller shall be able to bias one side faster/slower and stop one side when required.
FR-04	Avoid collision	Respond to insufficient	When distance falls below

		clearance.	the configured safety threshold, drive command shall transition to an avoidance state.
FR-05	Fail safe	Prevent uncontrolled movement on abnormal conditions.	Startup, sensor failure and battery/power faults shall lead to SAFE-STOP or controlled recovery.
FR-06	Race repeatability	Run the same control sequence predictably.	Use deterministic thresholds, bounded delays and a documented state machine.
NFR-01	Serviceability	Make faults easy to isolate.	Use labeled wiring, common ground rail, accessible connectors and modular electronics.
NFR-02	Packaging	Fit the competition envelope.	Target a compact rectangular chassis below 50 × 50 cm with sensor mounts inside the outline.
NFR-03	Electrical robustness	Avoid controller resets caused by motor transients.	Use separate regulated logic supply and sensible grounding/power distribution.

5. Exact Solution Architecture

5.1 Hardware Topology

The robot is a differential-drive 4WD platform. The Arduino Mega 2560 is the control core. Three HC-SR04 modules form a simple spatial sensing ring: FRONT, LEFT and RIGHT. The L298N provides two H-bridge channels; one channel is assigned to the left motor pair and the other to the right motor pair. Steering is achieved by creating a speed or direction difference between those two sides.

Subsystem	Component	Role
Controller	Arduino Mega 2560	Runs sensor acquisition, decision logic and motor commands.
Front sensing	HC-SR04 #1	Primary collision/clearance sensor.
Left sensing	HC-SR04 #2	Left-side clearance / wall context.
Right sensing	HC-SR04 #3	Right-side clearance / wall context.
Motor interface	L298N dual H-bridge	Converts controller IN/EN commands into power to the motor pairs.
Drive train	4 × 12V geared DC motors	Two motors mechanically placed on the left side, two on the right.
Power	Onboard battery + regulated 5V rail	Motor rail feeds L298N; regulated 5V rail feeds controller/sensors as appropriate.
Main isolation	SPST power switch	Allows safe energization/de-energization of the robot.

5.2 Electrical Pin Assignment

Signal	Mega pin	Function
FRONT TRIG	D30	HC-SR04 front
FRONT ECHO	D31	HC-SR04 front
LEFT TRIG	D32	HC-SR04 left
LEFT ECHO	D33	HC-SR04 left
RIGHT TRIG	D34	HC-SR04 right

RIGHT ECHO	D35	HC-SR04 right
L298N IN1	D24	Drive control
L298N IN2	D25	Drive control
L298N IN3	D26	Drive control
L298N IN4	D27	Drive control
L298N ENA	D5	Left-channel enable / PWM
L298N ENB	D6	Right-channel enable / PWM

5.3 Wiring Rules

- All grounds are common: Arduino, HC-SR04 modules, L298N logic/control ground and battery negative must share a defined ground path.
- Motor current must not be routed through the breadboard logic traces where a dedicated motor-power path is available.
- Use a regulated 5V supply for the Mega/sensors where the battery voltage is higher than 5V.
- Keep sensor signal wires away from high-current motor wires where practical, and secure loose leads so vibration cannot change the sensor orientation.
- Use the main switch to isolate the robot before connecting or disconnecting wiring.

5.4 Reference Electrical Schematic

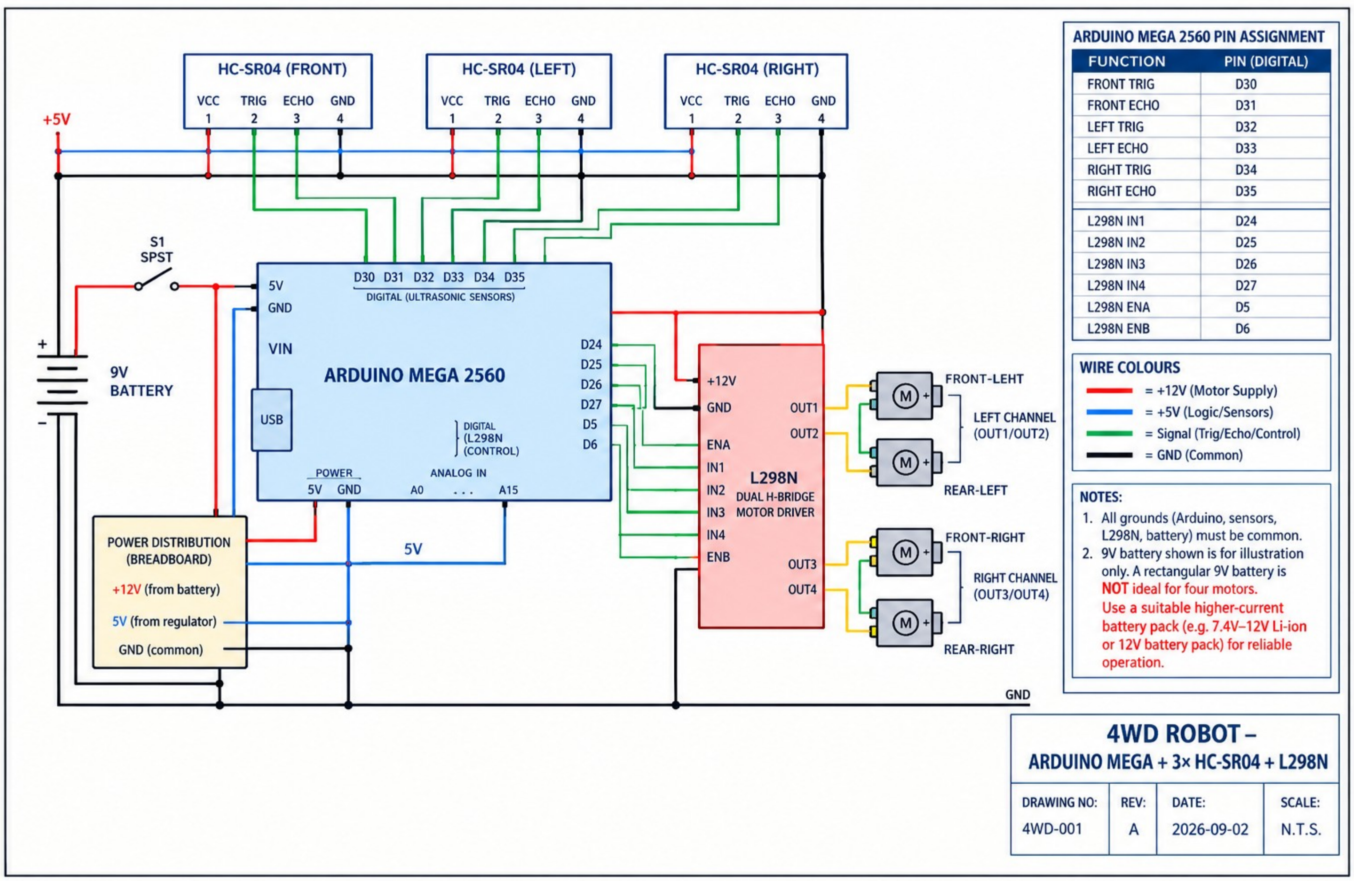


Figure 1 — Reference schematic. Treat the 9V PP3 as illustrative only; the finished four-motor platform needs a higher-current rechargeable pack.

6. Mechanical Design

6.1 Chassis Architecture

The mechanical design should use a rigid rectangular base with four driven wheels located symmetrically around the centerline. Heavy components such as the battery and motor driver should sit low and near the geometric center. The Arduino and sensor wiring can sit on an upper electronics deck provided the resulting center of gravity remains low enough to limit wheel lift during hard turns.

- Four-wheel drive provides traction and redundancy; left and right motor pairs act as logical drive channels.
- Sensor brackets should keep the three HC-SR04 faces square to their intended directions. Small angular errors can create uneven distance readings and false steering decisions.
- Protect electronics from wheel spray, loose wires and impact contact. A simple raised electronics plate is preferred to mounting modules directly against the floor.
- Fasteners should be mechanically locked against vibration where required. Cable ties or printed cable guides should restrain heavy motor leads.

6.2 Packaging and Envelope Control

Parameter	Design guidance
Maximum allowed footprint	500 × 500 mm
Target design margin	Keep the measured outer envelope comfortably below the limit instead of designing to the boundary.
Sensor extension	Sensor brackets and connector plugs count toward the practical envelope and should be included in inspection measurement.
Battery placement	Low and central to reduce pitch/roll moment during acceleration and avoidance.
Maintenance	Allow access to the main switch, battery connector, controller USB port and motor-driver terminals.

6.3 Drive Geometry and Steering

For differential steering, the robot yaw response depends on the speed difference between the left and right sides. A simple normalized control law is:

CONTROL LAW

left_cmd = base_speed + steer_bias; right_cmd = base_speed - steer_bias. Positive steer_bias produces a left/right speed difference; sign convention must be fixed during software commissioning.

The exact steering gain and base speed should be tuned experimentally because motor torque, wheel diameter, surface friction, battery voltage and chassis loading determine the actual turning response.

7. Sensing and Perception

7.1 HC-SR04 Operating Concept

Each HC-SR04 is activated by a trigger pulse and reports the time between the transmitted ultrasonic burst and the returning echo. Distance is calculated from the measured round-trip time and the speed of sound. A commonly used relationship is $d = v_{sound} \times t / 2$. At approximately 20 °C, v_{sound} is about 343 m/s.

For an autonomous racer, raw readings should not be used blindly. The controller should reject obviously invalid readings, apply a small median or moving-average filter where needed, and use thresholds with hysteresis so the robot does not chatter between two movement states.

7.2 Sensor Placement

Sensor	Purpose	Mounting	Control use
Front	Primary collision clearance	Forward-facing on centerline	Controls emergency braking / avoidance decision
Left	Left clearance / corridor context	Forward-left or left-facing	Biases steering away from close left boundary
Right	Right clearance / corridor context	Forward-right or right-facing	Biases steering away from close right boundary

7.3 Recommended Measurement Logic

1. Trigger one sensor and measure echo time.
2. Convert the valid time measurement to distance.
3. Reject readings outside the sensor's usable range or with a timeout.
4. Repeat for the other sensors using a schedule that prevents overlapping acoustic bursts.
5. Feed the filtered values into the decision state machine.
6. If the front distance falls below the emergency threshold, reduce base speed immediately and select the avoidance direction using side clearances.

IMPORTANT

The three ultrasonic sensors should not be fired simultaneously. Staggering measurements reduces the chance that one sensor hears another sensor's echo.

8. Motor Drive and Power System

8.1 Motor Driver Topology

The L298N is a dual H-bridge. In this four-motor implementation, the two left motors are paired on one bridge channel and the two right motors are paired on the second bridge channel. IN1/IN2 determine the left channel direction; IN3/IN4 determine the right channel direction; ENA and ENB enable the channels and can accept PWM for speed control.

8.2 Recommended Command Table

Action	Left channel	Right channel	Intent
FORWARD	IN1=1, IN2=0	IN3=1, IN4=0	Both sides forward
REVERSE	IN1=0, IN2=1	IN3=0, IN4=1	Both sides reverse
LEFT	Reduced left / stopped left	Right forward	Pivot or arc left, depending on tuning
RIGHT	Left forward	Reduced right / stopped right	Pivot or arc right, depending on tuning
BRAKE/STOP	0/0 or implementation-specific	0/0 or implementation-specific	Stop command; verify module behavior on the actual board

8.3 Battery and Power Budget

The procurement BOM contains a 9V PP3 battery because it is shown in the reference circuit. That battery is not an appropriate sustained source for four 12V geared motors. The competition build therefore requires a higher-current rechargeable battery pack that can supply the motor load and transient current without excessive voltage sag.

- Motor rail: choose a battery voltage compatible with the selected geared motors and the L298N operating range.
- Logic rail: derive a regulated 5V supply for the Arduino/sensors when the battery voltage is above 5V.
- Common ground: connect battery negative, driver ground and logic ground at a deliberate distribution point.
- Protection: include a master switch and, where practical, a fuse or current-limiting protection appropriate to the pack and wiring.

POWER WARNING

Do not treat the 9V PP3 as the final race battery for a four-motor drive train. The final hardware configuration must be verified for motor voltage, driver dissipation, current capability and battery discharge rating.

9. Autonomous Control Software

9.1 Software Architecture

Module	Responsibility
Initialization	Configure I/O, PWM channels, sensor timeouts and safety defaults.
Sensor service	Acquire front/left/right measurements on a staggered schedule.
Filtering	Validate and smooth readings; mark invalid data explicitly.
Decision engine	Convert distance context into a state such as DRIVE or EVADE.
Motion controller	Map the state into left/right motor direction and PWM commands.
Safety monitor	Stop or slow the robot when sensing fails, battery/logic resets occur or an invalid state is detected.
Telemetry/debug	During development only, expose readings and states through Serial; remove or minimize blocking debug output for final racing timing.

9.2 Suggested State Machine

State	Action	Entry condition
READY	Waiting for start condition; motors disabled.	Main switch on; controller healthy.
DRIVE	Normal forward racing.	No immediate obstacle and sensors valid.
SLOW	Reduce speed before close geometry.	Front distance entering warning band.
EVADE-LEFT	Bias or pivot left.	Left side offers more clearance than right, or route logic selects left.
EVADE-RIGHT	Bias or pivot right.	Right side offers more clearance than left, or route logic selects right.
RECOVER	Controlled re-acquisition of heading.	Robot has completed avoidance and needs to return to preferred trajectory.
SAFE-STOP	Motors disabled.	Sensor timeout, controller fault, emergency stop condition or shutdown.

9.3 Pseudocode

```
START
  initialize pins, PWM, sensors, safety defaults
  motorStop()
  state <- READY

LOOP
  front <- readFront()
  left  <- readLeft()
  right <- readRight()

  if invalid(front, left, right):
      state <- SAFE-STOP
  else if front < EMERGENCY_DISTANCE:
      if left > right: state <- EVADE-LEFT
      else: state <- EVADE-RIGHT
  else if front < WARNING_DISTANCE:
      state <- SLOW
  else:
      state <- DRIVE

  applyMotorCommand(state, left, right, front)
  waitUntilNextControlCycle()
```

10. Race Control and Tuning Strategy

10.1 Speed Strategy

A fast robot is not the same as a successful racer. The control strategy should trade speed for predictability near obstacles or tight sections of the course. A practical implementation uses at least two speed bands: a higher base speed in clear space and a reduced speed when the front sensor enters the warning zone.

10.2 Thresholds

Parameter	Purpose	Engineering basis
WARNING_DISTANCE	Start reducing speed	Tune from measured stopping distance + margin.
EMERGENCY_DISTANCE	Initiate decisive avoidance or stop	Must exceed the robot's worst-case stopping distance at the selected race speed.
SIDE_BIAS_THRESHOLD	Select left vs right path	Derived from difference between left and right measured clearance.
SENSOR_TIMEOUT	Declare sensing fault	Short enough to fail safe, long enough not to false-trigger on noisy returns.

10.3 Basic Stopping-Distance Method

For commissioning, measure actual stopping distance instead of assuming a theoretical value. Conduct repeat runs at the intended race speed, record the distance from obstacle detection to complete stop, and define the emergency threshold as the measured worst case plus a safety margin. This converts the avoidance logic from an arbitrary number into a verified engineering parameter.

11. Verification and Testing Plan

ID	Test	Method	Pass criterion
V-01	Power-up test	Switch on; Arduino starts; no	No unintended motion; logic

		motor movement at startup.	supply stable.
V-02	Sensor range test	Place target at known distances front/left/right.	Reported distance tracks target with acceptable repeatability.
V-03	Sensor cross-talk test	Trigger sensors in sequence with all three mounted.	No persistent false echoes caused by another sensor.
V-04	Motor direction test	Command each drive channel separately.	Left pair and right pair rotate in the documented directions.
V-05	Differential steering test	Run forward with fixed left/right PWM offsets.	Robot turns in the expected direction consistently.
V-06	Obstacle avoidance test	Drive toward obstacle at controlled speed.	Robot slows/turns before collision; repeated runs are consistent.
V-07	Full autonomous lap test	Run complete course with no manual intervention.	Completes the course within the intended safety and repeatability envelope.
V-08	Power endurance test	Operate at representative race load.	No controller resets, dangerous heating or loose connections.
V-09	Inspection test	Measure envelope and mass.	Footprint <= 50x50 cm and mass <= 5 kg.

12. Risk, Failure Modes and Mitigations

Failure mode	Effect	Risk	Mitigation
Motor current overload	L298N heating / voltage drop	High	Use suitable battery/driver thermal margin; test loaded drive; avoid stalled motors.
9V PP3 used in final build	Rapid voltage sag / motors unable to deliver torque	High	Replace reference battery with a suitable high-current rechargeable pack.
Sensor cross-talk	False distance reading	Medium	Stagger sensor triggering; filter invalid readings.
Loose motor wire	Intermittent drive / resets	Medium	Strain relief, locking connectors and post-vibration inspection.
Common-ground error	Erratic sensors / driver behavior	High	Use a defined common ground distribution point and continuity test.
Sensor misalignment	Robot steers unpredictably	Medium	Rigid brackets and alignment check during pre-run inspection.
Controller reset	Loss of autonomy	High	Separate logic power from motor rail; minimize electrical noise and brownout risk.
Excess speed for course	Overshoot / collision	Medium	Use speed bands and verify stopping distance empirically.

13. Bill of Materials and Cost Position

The separate procurement workbook prepared for this project totals the currently listed reference build at R4,432.43 including VAT. That total includes the 9V PP3 line shown in the reference circuit; the recommended higher-current final motor battery is excluded from the current total and must be budgeted separately.

Group	Qty	Component	Cost / note
Controller	1	Arduino Mega 2560	R1,364.79 incl. VAT (from

			current BOM)
Ultrasonic sensing	3	HC-SR04	R81.87 incl. VAT each (from current BOM)
Motor driver	1	L298N	R226.95 incl. VAT (from current BOM)
Drive motors	4	12V geared 65RPM motors	R571.38 incl. VAT each (from current BOM)
Reference power	1	9V PP3	Not suitable as the final motor battery
Other electrical/mechanical	Various	Switch, buck, breadboard, jumpers, brackets, clip	Included in BOM total

CURRENT BOM TOTAL

R4,432.43 incl. VAT — visually treated as a reference procurement total only. Final race cost should be updated after selecting the compliant high-current battery and any final chassis/manufacturing hardware.

14. Build, Assembly and Service Procedure

7. Assemble the chassis and verify wheel alignment without powering electronics.
8. Mount the four motors and confirm that all wheels spin freely and have the intended direction convention.
9. Install the L298N close to the motor wiring but protected from impact and conductive debris.
10. Mount the Arduino Mega and establish the regulated logic power rail.
11. Install and align the three HC-SR04 sensors with rigid brackets.
12. Create the common ground distribution and then route sensor/control signals separately from motor-current wiring.
13. Load the autonomous control firmware and execute startup, direction and sensor tests individually.
14. Only after bench verification, energize the complete system with the final competition battery.

14.1 Pre-Race Checklist

☐ Footprint measured below 50 × 50 cm.

☐ Complete robot mass measured below 5 kg.

☐ Battery securely mounted and isolated by main switch.

☐ No loose wires near wheels, gears or moving linkages.

☐ All three sensor brackets tight and correctly oriented.

☐ Motor direction verified.

☐ Sensor values verified before every run.

☐ Autonomy tested without manual steering.

☐ Emergency/stop procedure understood by the team.

15. Technical Rationale and Expected Performance

The main advantage of the selected architecture is simplicity. The Arduino Mega provides many digital I/O resources, the three ultrasonic modules are straightforward to integrate, and the L298N reduces the motor interface to two drive channels. This lowers software complexity and makes the first autonomous prototype easy to debug.

The main limitation is power and drive efficiency. The L298N introduces voltage loss and heat compared with newer MOSFET-based motor drivers, and the reference 9V PP3 cannot support sustained four-motor racing. Therefore, the performance envelope must be characterized experimentally rather than assumed. The highest-value improvements are a suitable high-current battery, robust power distribution, wheel alignment, sensor timing, and tuned differential steering.

15.1 Engineering Metrics to Record

ID	Metric	Unit	Measurement method
M-01	Top sustainable speed	m/s	10-run average on the actual race surface
M-02	Stopping distance	cm	Worst case at race speed
M-03	Sensor repeatability	cm	Standard deviation over repeated stationary tests
M-04	Motor-driver temperature rise	°C	After representative continuous drive
M-05	Runtime / course attempts	min or laps	Measure under representative battery condition
M-06	Mass	kg	Competition configuration
M-07	Footprint	mm × mm	Maximum external dimensions

16. Why This Fits Autonomous Racing

This platform is deliberately optimized around the key demands of autonomous racing: short decision cycles, simple but useful environmental perception, differential traction control, compact packaging and repeatable behavior. The robot does not depend on a human steering command during the run. Instead, the on-board controller closes the loop between measured distance and motor command.

RACING PRINCIPLE

Consistency beats theoretical peak speed. The preferred setup is the fastest configuration that can repeatedly complete the course without sensor-induced oscillation, motor-driver thermal problems or loss of traction.

17. Conclusion

The documented solution is a practical autonomous 4WD racer built around an Arduino Mega 2560, three ultrasonic sensors and a dual-channel motor driver. It meets the stated autonomous architecture requirements and provides a clear path from electrical integration to race tuning. The remaining engineering work is primarily validation: measure the complete robot, select and qualify the final high-current battery, tune the autonomous thresholds, and verify repeated full-course performance.

Revision A should be treated as the build-control baseline. Any change to controller, sensor count, motor driver, battery voltage, wheel diameter or pin assignment should trigger a controlled revision so that the wiring diagram, firmware and procurement list remain consistent.

Appendix A — Procurement Reference

The current BOM workbook should be treated as the procurement source of truth for component names, quantities, supplier references, stock codes and direct product URLs. Before purchase, verify live pricing, stock and the final battery specification.

Workbook: /mnt/data/4WD_Robot_Bill_of_Materials.xlsx